

R E V I E W

The prostate unit organizational model, from imaging to management: State of art

LUCIA FEDERICA CARPAGNANO¹, VALENTINA TESTINI², LAURA EUSEBI³,
FRANCESCO SAVERIO GUERRA⁴, RICCARDO GUGLIELMI⁵, ROBERTO BELLITTI²,
EMANUELE TATÒ¹, GIUSEPPE GUGLIELMI^{2,6}

¹Health Management, "Dimiccoli" Hospital, Barletta, Italy; ²Radiology Unit, "Dimiccoli" Hospital, Barletta, Italy; ³Radiology Unit, "Carlo Urbani" Hospital, Jesi, Italy; ⁴Radiology Unit, "Sacro Cuore di Gesù" Hospital, Gallipoli (LE), Italy; ⁵Department of Radiology and Nuclear Medicine, Luzerner Kantonsspital, Lucerne, Switzerland; ⁶Department of Clinical and Experimental Medicine, Foggia University School of Medicine, Foggia, Italy

ABSTRACT

Background and Aim: Prostate cancer represents one of the most common malignancies in men and a leading cause of cancer-related morbidity and mortality worldwide. In recent years, improvements in diagnostic imaging, biopsy techniques and therapeutic strategies have contributed to earlier detection and better risk stratification. At the same time, the growing complexity of prostate cancer management requires integrated and patient-centred organizational models. This narrative review aims to summarize the current evidence on the role of multiparametric magnetic resonance imaging (mpMRI) in prostate cancer diagnosis, active surveillance and treatment planning, and to describe the Prostate Unit as an innovative multidisciplinary organizational model starting from imaging.

Methods: We conducted a narrative review of the literature using PubMed and Google Scholar, primarily focusing on publications from 2010 onward. Inclusion criteria included international guidelines and consensus documents, with a focus on mpMRI, the Prostate Imaging Reporting and Data System (PI-RADS), and MRI-directed biopsy. Furthermore, we analyzed the organizational framework and clinical governance principles of the Prostate Unit model.

Results: Multiparametric MRI has become the reference imaging modality for the detection and local staging of clinically significant prostate cancer, allowing improved patient selection for biopsy and treatment. MRI-targeted biopsy increases diagnostic accuracy and reduces overdiagnosis compared with systematic transrectal ultrasound-guided biopsy. MpMRI also plays a central role in active surveillance protocols and therapeutic planning. The Prostate Unit model, based on multidisciplinary collaboration and integration of transversal services,



Received: 13 January 2026 | Accepted: 3 March 2026

Correspondence: Giuseppe Guglielmi, M.D., Professor of Radiology / Department of Clinical and Experimental Medicine, Foggia University School of Medicine, Viale L. Pinto, 1, 71121, Foggia, Italy / E-mail: giuseppe.guglielmi@unifg.it / ORCID: 0000-0002-4325-8330

enables coordinated patient management, optimization of resources, and improved appropriateness of care, with imaging representing the entry point of the diagnostic-therapeutic pathway.

Conclusions: The integration of advanced imaging techniques with a structured multidisciplinary organizational model is essential for contemporary prostate cancer management. The Prostate Unit represents an effective strategy to translate precision medicine into clinical practice, improving quality of care, patient outcomes and healthcare system efficiency. (www.actabiomedica.it)

Key words: prostate unit, prostate cancer, multiparametric magnetic resonance imaging, MRI-targeted biopsy, multidisciplinary team

Introduction

Prostate cancer is the second most common disease worldwide and the sixth major cause of death in males. It accounts for 14% of new cancer cases and 6% of total cancer deaths (1).

Men have the highest cancer incidence, accounting for 417,000 cases and 22.8% of all incidental cases in Europe (1).

The most popular methods for diagnosing prostate cancer are serum prostate-specific antigen (PSA) levels (2).

When prostate cancer is suspected due to increased PSA levels and an abnormal digital rectal examination (DRE), a transrectal ultrasonography (TRUS) - guided biopsy is required to confirm the diagnosis.

Trans-rectal ultrasonography (TRUS)-guided systematic biopsy (SBx) is typically the following diagnostic step. However, TRUS has a limited role in detecting PCa because the majority of hypoechoic lesions are benign, with up to 50% of palpable tumours being invisible on imaging (2).

Magnetic resonance imaging (MRI) is the gold standard technique for diagnosing clinically relevant prostate cancer and staging tumours. Another important application is the introduction of MR-compatible devices, which enable invasive operations such as targeted prostate biopsy to be conducted.

Prostate cancer mortality has been decreasing as early detection and treatment procedures have improved (1).

Prostate cancer patients' prognosis is determined by parameters such as disease severity, histological aggressiveness, Gleason score, and PSA (2).

With these premises, for the diagnosis and treatment of prostate disease, it is essential to implement a new organizational model that allows the creation of the "Prostate Unit" area with the following objectives: implement multidisciplinary management; achieve early diagnosis of the disease; build patient loyalty, who must be able to count on a comprehensive treatment center, guarantee a multidisciplinary approach to treatment and a Unit Care organizational model; ensure cutting-edge surgery with the application of traditional and innovative methods in relation to the clinical evaluation of the individual, allowing the implementation of robotic surgery as a precision medicine technique in the oncology field; outline a path to humanize treatment, also dedicating greater attention and psychological support to patients affected by oncological diseases; offer remote services through the smart Telemedicine mode; carry out "second opinion" consultations with other treatment centers; optimize all resources that can "bridge" effectively contributing to the treatment path with attention to aspects that are collateral to diagnosis and therapy but relevant to the clinical outcome (clinical psychology, clinical nutrition, laboratory services, rehabilitation), creating an integrated platform of services that are transversal to the Operating Units; take care of internal and external communication aspects and in particular health communication between professionals, between professionals and patients, caregivers, companies and patient associations, external users;

train healthcare personnel, focusing on new hires with valid training and innovative research support.

This article aims to provide an overview of multiparametric MRI and evaluate its emerging role in three challenging areas of prostate cancer management: early diagnosis, active surveillance and treatment planning; moreover, starting from imaging, the authors propose a multidisciplinary management project that interconnects the different protagonists of the patient's path, in order to manage the patient in the Prostate Unit.

We conducted a narrative review of the literature using PubMed and Google Scholar as databases, focusing primarily on publications from 2010 to the present, using keywords such as "prostate unit, prostate cancer, multiparametric magnetic resonance imaging, targeted biopsy."

Inclusion criteria included international guidelines and consensus documents, with a focus on mpMRI, the Prostate Imaging Reporting and Data System (PI-RADS), and MRI-guided biopsy.

Imaging

Conventional US

TRUS is the most often used technology for prostate evaluation because of its great image resolution. It offers clear pictures of the gland, its borders, surrounding bladder, seminal vesicles, and anterior rectal wall.

TRUS is used mostly to calculate prostate volume. TRUS can also be used to evaluate the seminal vesicles, vas deferens, bladder, distal ureter, and urethra.

In a normal prostate, the outside zone is strongly echogenic and clearly distinguishable from the core gland, which is often hypoechoic and heterogeneous owing to BPH. Prostate cancer is distinguished by a circular or oval centre of low echogenicity in the periphery zone, which may be isoechoic or hyperechoic (3).

In compared to the peripheral zone, around 37-50% of tumours are isoechoic or slightly hypoechoic (4). When evaluating a low echoic focal lesion, the differential diagnosis is inflammation, fibrosis, infarction, or benign prostatic hyperplasia nodules (4).

B-mode TRUS has a limited capacity to identify tumours in the central glands. According to Onur et al.

(5) of 3,912 consecutive patients who received TRUS-guided biopsies, 25.5% had a hypoechoic lesion and 25.4% did not.

Multiparametric MRI

To identify potential prostate lesions, multiparametric MRI combines many imaging modalities, including T2WI, DWI, and DCE.

Multiparametric MRI includes T2-weighted imaging, DWI, and DCE, and has been shown to increase MRI's predictive potential for prostate cancer detection and staging (6).

MRI using various anatomical sequences has been shown to be effective for preoperative detection and local staging of prostate cancer in many studies. MRI is useful for the local staging before treatment, but now with the use of multiparametric protocol, MRI has shown accurate tumor detection with high sensitivity and specificity (7).

As part of its prostate imaging MRI recommendations, the European Society of Urogenital Radiology (ESUR) created the Prostate Imaging Reporting and Data System (PIRADS) in 2012, based on expert consensus to standardise the interpretation and reporting of prostate MRI (8). The authors developed a defined approach for standardizing the techniques of mpMRI acquisition, reporting and communication of results in order to detect prostate cancer.

The PIRADS document represents the current standard for image acquisition and interpretation based on expert consensus. The accuracy and reproducibility of this first generation system (sensitivity and specificity of 0.78 and 0.79, respectively) has been validated in a meta-analysis (8). Although PI-RADS provided a common reporting language, several issues remained. The framework of PIRADS is designed to help radiologists decide whether a lesion is likely to represent clinically significant cancer or not, using a scoring system of 1-5 (9).

Based on PI-RADS criteria, each lesion suspicious for tumour inside the prostate is assigned a point value between 1 and 5 for mpMRI, which commonly includes T2W, DWI, and DCE.

A score of 1 suggests that a lesion is most likely benign, whereas a score of 5 indicates a high risk of

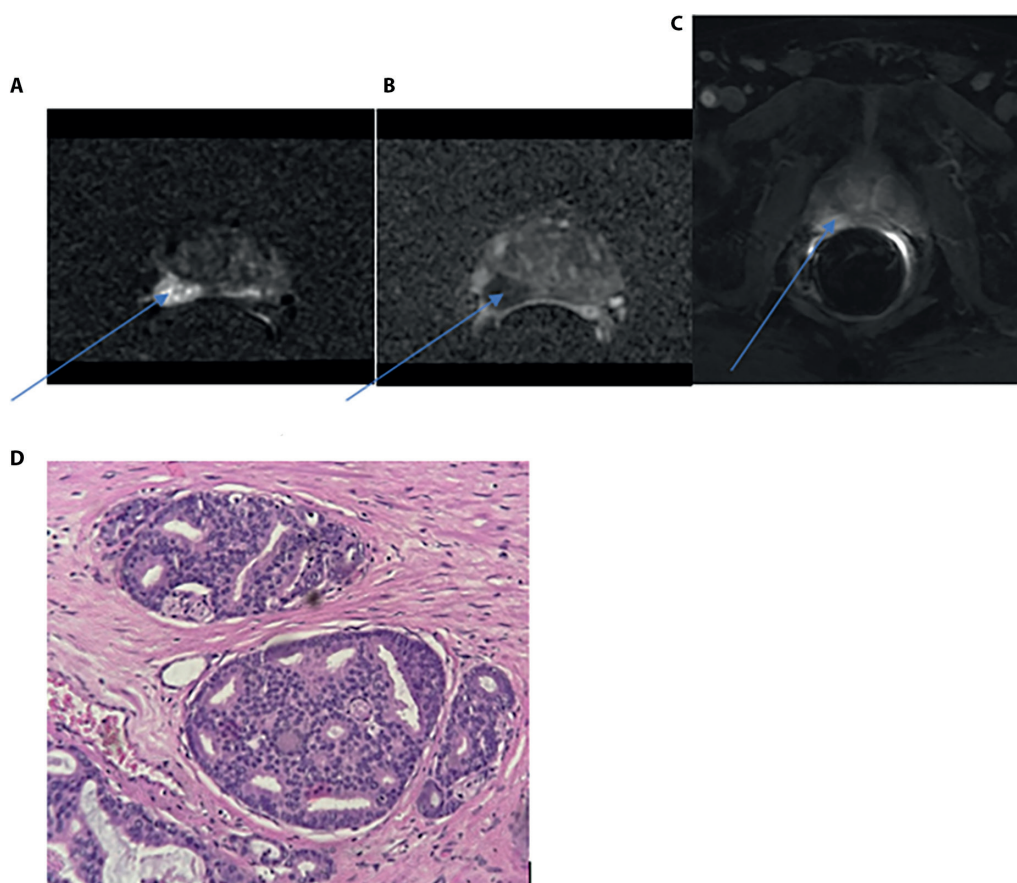


Figure 1. DWI sequences A), ADC maps B), and T1w imaging after contrast administration C) of a case of a prostate tumor. The arrow shows a big high signal lesion on DWI (A) with low signal in ADC B) and intense contrast enhancement C): this is a case of a PI-RADS 5. The histological examination of the lesion D) confirmed the presence of a malignant tumor with a Gleason score of 4+4.

malignancy. Every lesion considered to be a tumour is awarded a total point value (10). (Figure 1) In 2019, they upgraded PI-RADS to version 2. PI-RADS v1 and v2 have different principal determining sequences dependent on zone. According to PI-RADS v2, the primary establishing sequence for the peripheral zone is DWI, whereas the major establishing sequence for the transition zone is T2-weighted imaging (8).

The role of prostate biopsy

Screening with PSA and DRE is still important for risk assessment and early detection, but improved

diagnostic procedures and MRI-directed targeting have reduced overdiagnosis and overtreatment. (11).

According to EAU guidelines, prostate biopsy is the gold standard for PC diagnosis. The first diagnostic biopsy method is the random procedure (rather than sextant), in which 12 samples are collected only from the peripheral zone using transrectal ultrasound (TRUS)-guided prostate biopsy method (12). This technique has limitations in undersampling (insufficient number of samples in relation to prostate volume), oversampling (excessive core number, resulting in the identification of non-significant microfoci), overstaging, with the diagnosis of non-significant tumours (low sensitivity), understaging, with the failure

to diagnose clinically significant cancers (low specificity), and operator experience-related errors (13).

The transition from ten-to-12-core transrectal ultrasound (TRUS)-guided prostate biopsy to MRI-directed targeting plus systematic transperineal prostate biopsy has resulted in safer biopsy procedures, improved diagnoses, and increased accuracy (11).

PSA screening is an important risk-assessment tool in triaging select men for additional investigation, although it is not recommended or regularly conducted in all asymptomatic men (11,14).

Inappropriate use of PSA screening can result in unnecessary biopsies and overtreatment of benign conditions such as benign prostatic hyperplasia, prostatitis, and manipulation of the prostate.

Early detection of prostate cancer: A close relationship between anamnesis, laboratory data and imaging

The US Preventive Services Task Force (USPSTF) updated their guidelines in 2018, citing the availability of longer follow-up data from large screening studies, as well as increasing evidence that active monitoring in low-risk prostate cancer lowers the harms associated with screening overtreatment. The USPSTF currently recommends that men aged 55-69 years undergo PSA testing after talking with their doctor about the relative benefits and risks, but it continues to advise against PSA screening in younger men (40-55 years) and those over 70 years (15).

In the absence of broad, coordinated PSA testing, opportunistic testing has become commonplace in numerous EU member states. However, a new study suggests that, while this approach has minimal influence on prostate cancer-specific mortality, it is associated with more overdiagnosis than organised risk-adapted PSA testing (16,17).

Overdiagnosis can be reduced by utilising a risk-adapted detection technique based on PSA levels, together with risk calculators and multiparametric magnetic resonance imaging (mpMRI), which can distinguish between serious and insignificant prostate cancer and alter medication appropriately (18). As a

result, many early prostate cancer diagnoses may be handled with active surveillance, avoiding overtreatment (19). Those with a lower risk profile may benefit from local therapy, which has fewer side effects and has better results than detecting and treating the condition later, perhaps increasing or preserving the patient's quality of life (20).

The EAU created an algorithm to describe a risk-adapted approach for prostate cancer diagnosis. This approach is intended for usage by well-informed males over the age of 50 with a life expectancy of more than 10-15 years. This algorithm clearly demonstrates how to detect serious prostate cancer at an early stage while avoiding unnecessary diagnosis and treatment. Following a clinical risk assessment and appropriate counselling, the PSA test is the first step in identifying a large proportion of men with a low PSA value who do not require any further immediate investigations for 2-4 years (for those with a PSA value of 1-3 ng/ml) or 5 years (for those with a PSA value of 1 ng/ml and 60 years old) (19).

A risk stratification nomogram (which takes into account factors such as age, family history, digital rectal examination, and prostate volume [PSA density] in a risk calculator) will identify a subset of men (approximately 35% of all men with an initial PSA test of >3 ng/ml (21)) as having a low risk and requiring only clinical follow-up, avoiding the need for additional testing such as an MRI and biopsy. Men with a PSA value of more than 3 ng/ml who were classified as intermediate or high risk would then have mpMRI, resulting in the identification of a further subgroup (approximately 54% of all men undergoing MRI (21)) with a Prostate Imaging Reporting and Data System (PIRADS) score of 1-2 considered to be at a low risk of having significant prostate cancer and requiring clinical follow-up only.

MRI is an optimal method for guiding biopsy in men with a previous negative TRUS-guided biopsy. In fact, the false-negative rate of TRUS guided biopsies is 30-40 %; therefore, when biopsy is negative but a rise in PSA justifies further investigation, MRI should be used to detect areas suspicious for cancer to guide the biopsy (20).

MRI is feasible also in active surveillance for men with low-risk prostate cancer. Lymph node staging of prostate cancer using MRI is unreliable, as metastatic

lymph nodes in prostate cancer are often small, and conventional size criteria may underestimate the extent of nodal disease (21).

Prostate cancer typically manifests as focal not well-defined, low-signal-intensity area in the peripheral zone. Tumors localized to the transition zone or central gland are more challenging to detect since the signal intensity characteristics of the normal and hypertrophic transition zone usually overlap with tumor (22).

Signal heterogeneity of the central zone poses additional difficulties in the interpretation of MR images of the prostate since the stromal type of benign prostatic hyperplasia has low signal intensity similar to the prostate cancer. Findings that may support the diagnosis of cancer are lack of the low-intensity rim usually seen in case of benign prostatic hyperplasia, homogenous low signal intensity, ill-defined margins, lenticular shape, low signal intensity of the retroprostatic fat and fibromuscular stroma invasion (7).

T2-weighted imaging, despite the aforementioned limitations, is the basic tool to detect extracapsular extension (ECE) of prostate cancer. Findings which predict local advanced disease are bulging prostatic contour, irregular margin, direct breach of the capsule, capsular retraction, obliteration of the rectoprostatic angle, and asymmetry or encasement of the neurovascular bundle. In case of seminal vesicle invasion (SVI), findings include enlarged low signal seminal vesicles, obliteration of the angle between prostate and seminal vesicles, and direct extension. Irregular contour of the prostate gland, low signal intensity of the retroprostatic fat, and direct breach of the capsule are the criteria with the highest sensitivity, while the specificity is high for all of the abovementioned. (7,21)

Assessment of ECE and SVI in case of prostate cancer is crucial to determine the proper therapeutic approach, since ECE and SV invasion are very important prognostic factors for recurrence after radical prostatectomy. Patients with advanced local disease by the means of ECE and SV invasion have overall worse prognosis since there is greater risk of a positive surgical margin and increased incidence of lymph node metastasis.

Lymph node involvement in prostate cancer patients is a question since assessing node size and

morphology to predict lymph node neoplastic disease is inaccurate. Normal-sized lymph nodes may hide metastatic disease impossible to reveal using only morphologic criteria (7,21).

Diagnostic pathway for individuals suspected for prostate cancer

Adult men, with suspicion of prostate cancer based on a PSA increase or clinical suspicion on rectal examination, are indicated to perform a biopsy and to perform diagnostic investigation by means of MRI examination.

The AIOM guidelines report that 1) mpMRI is strongly indicated in naive patients with clinical suspicion of prostate cancer, in order to possibly guide a subsequent biopsy sampling and at the same time provide a local staging; in the case of a negative mpMRI and high clinical suspicion, the biopsy is still indicated and 2) mpMRI is strongly indicated in patients with a negative biopsy and persistent clinical suspicion (23).

In the event that the histological examination confirms the suspicion of malignancy, the patient will be a candidate for surgery or radiotherapy treatment (11).

We conducted a narrative review of the literature using PubMed and Google Scholar, primarily focusing on publications from 2010 onward. Inclusion criteria included international guidelines and consensus documents, with a focus on mpMRI, the Prostate Imaging Reporting and Data System (PI-RADS), and MRI-directed biopsy. Furthermore, we analyzed the organizational framework and clinical governance principles of the Prostate Unit model, including a real-world institutional experience from the hospital where we work.

Active surveillance

Active surveillance (AS) can be proposed only to selected patients (24,25):

- Very low risk of progression
- Life expectancy greater than 10 years

- Clinical stage T1-2
- Gleason score 6
- PSA ≤ 10 ng/ml
- ≤ 2 biopsy cores involved by carcinoma.

The decision to perform an AS must be shared with the patient, after adequate information and expression of consent, evaluating the performance of a mpMRI in the case of more than two positive cores. The aim of the AS is to avoid or delay treatment with the aim of radicality and healing, keeping the patient under strict clinical and biochemical control and repeating prostate biopsies at pre-established intervals (11). A possible radical treatment is proposed when there is reasonable certainty that the disease may become clinically significant (increase in PSA, increase in Gleason score/increase in the number of positive samples for neoplasia on re-biopsy), or if the patient expresses the will to exit the SA program (24,25).

The follow-up schedule in the AS regimens includes:

- Determination of PSA every 3 months,
- Clinical control every 6 months,
- Saturation re-biopsies, after an mpMRI.

Prostate unit and patient management

The “PROSTATE UNIT” project is part of a complex context made up of regulatory requirements, ethics, law and scientific clinical research.

The necessary experimental organizational model meets all the requirements of the 8 Institutional Accreditation Requirements and creates a bridge between the hospital and the territory, in compliance with the provisions of Ministerial Decree 77 and the PNRR.

The Gantt Diagram shows the implementation phases of the experimental project according to the Deming cycle (Figure 2). It is a project shared with clinicians dedicated to the care of patients suffering from oncological pathologies, which must then be included in the Budget Document in order to strengthen and

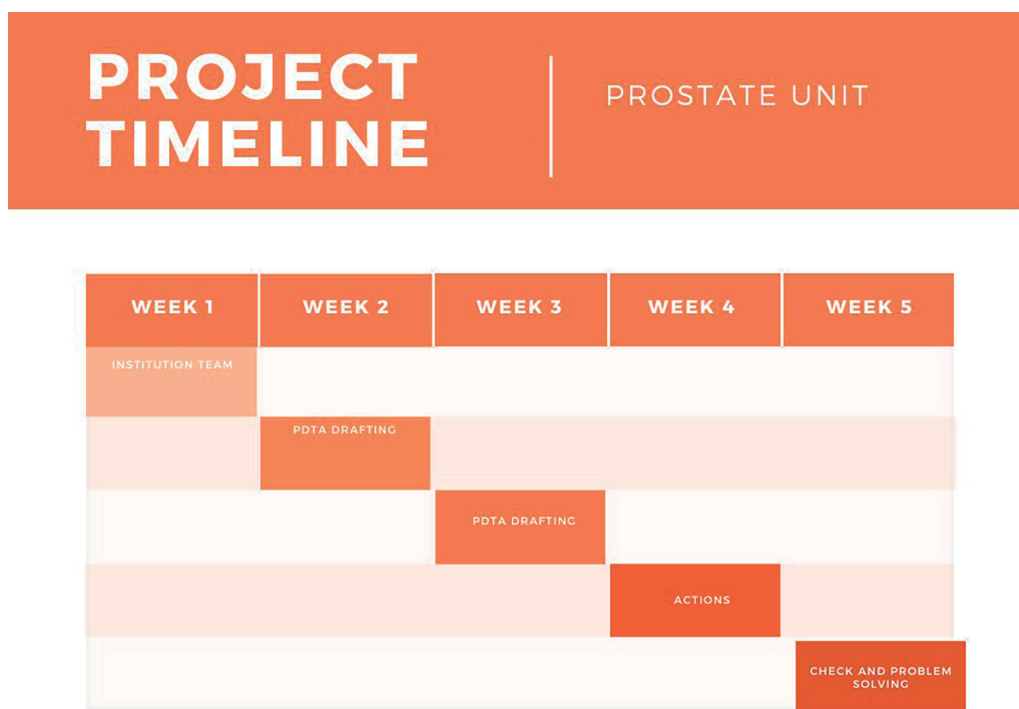


Figure 2. The Gantt Diagram shows the implementation phases of the experimental project according to the Deming cycle.

root the objectives of the model itself in a Management System, with a formal deliberative act (26,27).

Prostate unit: our experience of ASL BT

The Italian Government has recently established that oncological pathologies must represent a priority, in terms of research and clinical assistance, highlighting the need for a strategic plan focused on patients suffering from oncological pathologies, in order to improve resources and opportunities for them.

In local implementation, the Strategic and General Management of the Barletta-Andria-Trani (BT) Local Health Authority (ASL) promoted and definitively approved the ASL BT Oncology Network Organizational Model with the initiative of creating a specific multidisciplinary group for each oncological pathology (Multidisciplinary Team). The teamwork model between different specialists is not new in the field of oncology, which is characterized by a transversal approach by definition. As a novelty compared to the traditional multidisciplinary approach, focused on a single disease or an organ that is still systemically dependent, we have integrated into the Teams all the specialists who are most likely to intercept patients with clinical manifestations attributable to prostate cancer. The specialists included in the Team are not intended as a closed list, but a core group of doctors who actively contribute to the recognition and management of prostate cancer.

The first very practical objective of establishing the Team was to increase doctors' knowledge and awareness of multidisciplinary in uroradiology and to implement intra-Unit and inter-Unit referral by identifying a targeted specialist in each Unit participating in the Team. Multidisciplinary discussion takes place in different formats, including patient assessment in the presence of several specialists together, or focused discussions on difficult cases between specialists only. However, an approach based on the red flag system and a roadmap shared by all specialists allows the participation of different doctors in the multidisciplinary discussion to be personalized and optimized on a single case basis, avoiding recruiting the entire Group for each case but at the same time time to provide patients with the necessary expertise. On the other hand, the

shared system harmonizes the specialists in the first approach and saves the patient time on the path to final confirmation of the diagnosis.

Once the single-specialist evaluation has been completed, the patient is evaluated at the multidisciplinary clinic hosted at the ASL BT and the case is further discussed with the treating doctor, the oncologist, the radiotherapist and other specialists of the Group, or external ones, if requested. Once the diagnosis is confirmed, the therapeutic approach is evaluated; if necessary, off-label or non-marketed options are explored together with the Pharmacy Unit. Follow-up visits are scheduled and each case is regularly discussed again following the evolution of the disease and the response to therapy.

Radiology represents the patient's entry point into the organizational clinical path which requires timely and adequate management.

Imaging is the initial phase of a path that is born and grows with professional radiological techniques and skills, to then always culminate with imaging, in the monitoring of specific therapeutic diagnostic parameters

The Multidisciplinary Team can also count on the skills of "transversal services", including Laboratory, Pharmacy, Clinical Psychology, Nutrition, Physiotherapy, Pathological Anatomy, Radiology, which acts as a common basis for all the Units and Multidisciplinary Groups that manage oncological pathologies in within the ASL BT Oncology Network Organizational Model.

The production cycle of taking charge is shown in Figure 3.

Discussion

From a regulatory and contextual analysis of the roles that Accreditation, Quality and Safety Systems, and Clinical Risk play in the new Healthcare Organization landscape, the increasingly strategic role of a Healthcare Management team emerges. This management team, in accordance with the principles of Clinical Governance, is capable of applying organizational methodologies aimed at ensuring improved quality of clinical care for patients, as well as safety and

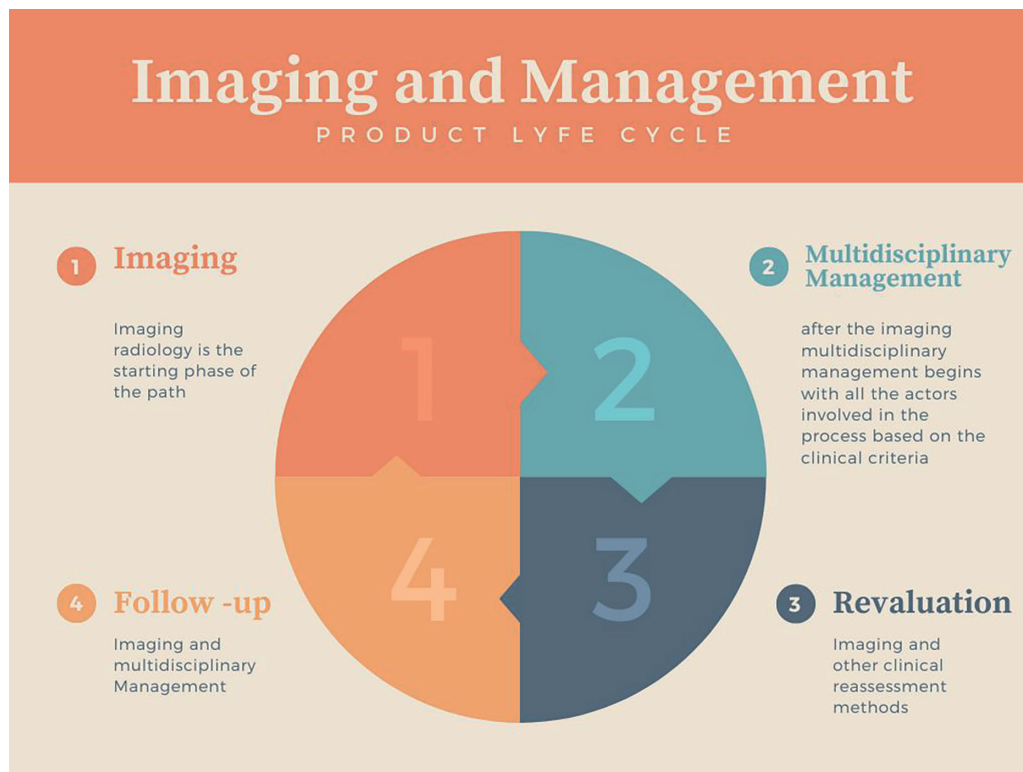


Figure 3. The production cycle of taking charge.

well-being in the working environment for all healthcare workers. (28,29)

This review aims to define a set of monitoring, process, and outcome indicators within the healthcare organization. These can provide a fertile ground for Accreditation Institutions and Bodies to standardize care and reception pathways, maximizing the potential of all clinical and healthcare centers to deliver their full potential, thus optimizing resources and pursuing the goal of Health Equality.

Our Prostate Unit experience constitutes an experimental model, not only of healthcare management, but also of training and research in Healthcare Management, whose guiding principle is “From Research to Healthcare Management.”

Conclusions

In recent decades, several medical specialties have undergone a revolution, following the approach

of so-called precision medicine, based on innovative diagnostic techniques and new targeted drugs. In this context, developing an innovative organizational methodology is crucial. To offer all patients appropriate diagnosis and treatment options, the structure of care pathways must be rethought accordingly.

For this reason, it is essential to guarantee patients multidisciplinary management, such as the Prostate Unit, to interconnect the various actors in the patient journey into a network, starting with imaging as the initial phase.

The Clinical Governance model we have adopted, applied experimentally to the Prostate Unit, is based on an Accreditation Manual for organizational processes, with the following core elements: Accreditation and standards of excellence; Telemedicine, Robotic Surgery, Precision Medicine, Communication and empowerment of patients, caregivers, and associations, Multidisciplinary management, Second opinion, Resource optimization through the sharing of innovative specialists, Ultra-specialized professional training,

Waiting List Management, and Fast Track Consultation. This is an innovative strategic organizational model: “from research to healthcare management,” which addresses healthcare needs in the areas of Management, Oncology, Robotic Surgery, and Highly Specialized Radiology.

Ethic Approval: Not applicable.

Conflict of Interest: Each author declares that he or she has no commercial associations (e.g., consultancies, stock ownership, equity interest, patent/licensing arrangement, etc.) that might pose a conflict of interest in connection with the submitted article.

Author's Contributions: All Authors contributed to the article's conception and design. All authors participated in the writing of the paper. All authors read and approved the final manuscript. The Authors have no financial or competing interests to disclose.

Declaration on the use of AI: None.

Informed consent statement: Not applicable.

Acknowledgments: None.

Funding: The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

References

1. Ferlay J, Steliarova-Foucher E, Lortet-Tieulent J, et al. Cancer incidence and mortality patterns in Europe: estimates for 40 countries in 2012. *Eur J Cancer*. 2013;49(6):1374–403. doi: 10.1016/j.ejca.2012.12.027
2. Mowatt G, Scotland G, Boachie C, et al. The diagnostic accuracy and cost-effectiveness of magnetic resonance spectroscopy and enhanced magnetic resonance imaging techniques in aiding the localisation of prostate abnormalities for biopsy: a systematic review and economic evaluation. *Health Technol Assess*. 2013;17(20). doi: 10.3310/hta17200
3. Vo T, Rifkin MD, Peters TL. Should ultrasound criteria of the prostate be redefined to better evaluate when and where to biopsy? *Ultrasound Q*. 2001;17(3):171–6. doi: 10.1097/00013644-200109000-00005
4. Ismail M, Gomella LG. Ultrasound for prostate imaging and biopsy. *Curr Opin Urol*. 2001;11(5):471–7. doi: 10.1097/00042307-200109000-00004
5. Onur R, Littrup PJ, Pontes JE, et al. Contemporary impact of transrectal ultrasound lesions for prostate cancer detection. *J Urol*. 2004;172(2):512–4. doi: 10.1097/01.ju.0000131621.61732.6b
6. Hegde JV, Mulkern RV, Panych LP, et al. Multiparametric MRI of prostate cancer: an update on state-of-the-art techniques and their performance in detecting and localizing prostate cancer. *J Magn Reson Imaging*. 2013;37(5):1035–54. doi: 10.1002/jmri.23860
7. Vargas HA, Hötter AM, Goldman DA, et al. Updated Prostate Imaging Reporting and Data System (PI-RADS v2) recommendations for the detection of clinically significant prostate cancer using multiparametric MRI: critical evaluation using whole-mount pathology as standard of reference. *Eur Radiol*. 2016;26(6):1606–12. doi: 10.1007/s00330-015-4015-6
8. Hamoen EHJ, de Rooij M, Witjes JA, et al. Use of the Prostate Imaging Reporting and Data System (PI-RADS) for prostate cancer detection with multiparametric magnetic resonance imaging: a diagnostic meta-analysis. *Eur Urol*. 2015;67(6):1112–21. doi: 10.1016/j.eururo.2014.10.033
9. Abd-Alazeez M, Ahmed HU, Arya M, et al. Can multiparametric magnetic resonance imaging predict upgrading of transrectal ultrasound biopsy results at more definitive histology? *Urol Oncol*. 2014;32(6):741–7. doi: 10.1016/j.urolonc.2014.01.008
10. Röhke M, Blondin D, Schlemmer HP, et al. PI-RADS-Klassifikation: Strukturiertes Befundungsschema für die MRT der Prostata [PI-RADS classification: structured reporting for MRI of the prostate]. *Radio*. 2013;185(3):253–61. doi: 10.1055/s-0032-1330270
11. European Association of Urology. EAU Guidelines. Arnhem, The Netherlands: European Association of Urology; 2022. Available from: <https://uroweb.org/guidelines>. Accessed April 2022.
12. Siddiqui MM, Rais-Bahrami S, Truong H, et al. Magnetic resonance imaging/ultrasound-fusion biopsy significantly upgrades prostate cancer versus systematic 12-core transrectal ultrasound biopsy. *Eur Urol*. 2013;64(5):713–9. doi: 10.1016/j.eururo.2013.05.059
13. Bjurlin MA, Meng X, Le Nobin J, et al. Optimization of prostate biopsy: the role of magnetic resonance imaging targeted biopsy in detection, localization and risk assessment. *J Urol*. 2014;192(3):648–58. doi: 10.1016/j.juro.2014.03.117
14. National Health and Medical Research Council. Prostate-specific antigen (PSA) testing in asymptomatic men: evidence evaluation report. Canberra: NHMRC; 2013. Available from: <https://www.nhmrc.gov.au/>. Accessed April 2022.

15. US Preventive Services Task Force, Grossman DC, Curry SJ, Owens DK, et al. Screening for prostate cancer: US Preventive Services Task Force recommendation statement. *JAMA*. 2018;319(18):1901-13. doi: 10.1001/jama.2018.3710
16. Arnsrud Godtman R, Holmberg E, Lilja H, et al. Opportunistic testing versus organized prostate-specific antigen screening: outcome after 18 years in the Göteborg randomized population-based prostate cancer screening trial. *Eur Urol*. 2015;68(3):354-60. doi: 10.1016/j.eururo.2014.12.006
17. Engler J, Dahlhaus A, Güthlin C. The readiness of German GPs to recommend and conduct cancer screening is associated with patient-physician gender concordance: results of a survey. *Eur J Gen Pract*. 2017;23(1):11-9. doi: 10.1080/13814788.2016.1240166
18. Nordström T, Aly M, Clements MS, et al. Prostate-specific antigen (PSA) testing is prevalent and increasing in Stockholm County, Sweden, despite no recommendations for PSA screening: results from a population-based study, 2003-2011. *Eur Urol*. 2013;63(3):419-25. doi: 10.1016/j.eururo.2012.10.001
19. Van Poppel H, Roobol MJ, Chapple CR, et al. Prostate-specific antigen testing as part of a risk-adapted early detection strategy for prostate cancer: European Association of Urology position and recommendations for 2021. *Eur Urol*. 2021;80(6):703-11. doi: 10.1016/j.eururo.2021.07.024
20. Scattoni V, Zlotta A, Montironi R, et al. Extended and saturation prostatic biopsy in the diagnosis and characterization of prostate cancer: a critical analysis of the literature. *Eur Urol*. 2007;52(5):1309-22. doi: 10.1016/j.eururo.2007.08.006
21. Turkbey B, Rosenkrantz AB, Haider MA, et al. Prostate Imaging Reporting and Data System version 2.1: 2019 update of Prostate Imaging Reporting and Data System version 2. *Eur Urol*. 2019;76(3):340-51. doi: 10.1016/j.eururo.2019.02.033
22. Carpagnano FA, Eusebi L, Tupputi U, et al. Multiparametric MRI: local staging of prostate cancer. *Curr Radiol Rep*. 2020;8:27. doi: 10.1007/s40134-020-00352-1
23. Associazione Italiana di Oncologia Medica (AIOM). Linee guida: Carcinoma della prostata. Edizione 2024. Milano: AIOM; 2024. Available from: <https://www.aiom.it/linee-guida-aiom/>. Accessed July 2026
24. Willemse PM, Davis NF, Grivas N, et al. Systematic review of active surveillance for clinically localised prostate cancer to develop recommendations regarding inclusion of intermediate-risk disease, biopsy characteristics at inclusion and monitoring, and surveillance repeat biopsy strategy. *Eur Urol*. 2022;81(4):337-46. doi: 10.1016/j.eururo.2021.12.007
25. Mottet N, Van den Bergh RCN, Briers E, et al. EAU-EANM-ESTRO-ESUR-SIOG guidelines on prostate cancer—2020 update. Part 1: screening, diagnosis, and local treatment with curative intent. *Eur Urol*. 2021;79(2):243-62. doi: 10.1016/j.eururo.2020.09.042
26. Blendon RJ, Benson JM. Americans' views on health policy: a fifty-year historical perspective. *Health Aff (Millwood)*. 2001;20(2):33-46. doi: 10.1377/hlthaff.20.2.33
27. Clancy CM, Glied SA, Lurie N. From research to health policy impact. *Health Serv Res*. 2012;47(1 Pt 2):337-43. doi: 10.1111/j.1475-6773.2011.01374.x
28. Heidenreich A, Bellmunt J, Bolla M, et al. EAU guidelines on prostate cancer. Part 1: screening, diagnosis, and treatment of clinically localised disease. *Eur Urol*. 2011;59(1):61-71. doi: 10.1016/j.eururo.2010.10.039
29. Fondazione GIMBE. Il governo clinico nelle aziende sanitarie. Position statement. Bologna: Fondazione GIMBE. Available from: <https://www.gimbe.org/>.

Copyright: The Author(s), 2026. Licensee Mattioli 1885, Fidenza, Italy. This is an open-access article distributed under the terms of the Creative Commons Attribution NonCommercial License (CC BY-NC-4.0).

Disclaimer/Publisher's Note: The statements, opinions and data contained in this article are solely those of the author(s) and contributor(s) and do not necessarily reflect those of their affiliated organizations, the publisher, the editors or the reviewers. The publisher and the editors disclaim any responsibility for injury to people or property resulting from any ideas, methods, instructions or products mentioned in the content. Any product that may be evaluated in this article, or claim made by its manufacturer, is not guaranteed or endorsed by the publisher.